It will decrease as the temperature increases. A high viscosity fuel may cause extreme pressures in the injection systems and will cause reduced atomization and vaporization of the fuel spray. The viscosity of diesel fuel must be low enough to flow freely at its lowest operational temperature, yet high enough to provide lubrication to the moving parts of the finely machined injectors. The fuel must also be sufficiently viscous so that leakage at the pump plungers and dribbling at the injectors will not occur. Viscosity also will determine the size of the fuel droplets, which, in turn, govern the atomization and penetration qualities of the fuel injector spray.

5. **Sulphur Content:** The sulphur in fuel will cause wear of the internal components of the engine, such as piston ring, pistons, valves, and cylinder liners. In addition, a high sulphur content fuel requires that the engine oil and filter be changed more often. This is because of formation of acids when sulphur-di-oxide formed during combustion combines with water vapor. The Sulphur content should be less than 0.5%
6. **Moisture and Sediment Content:** Cleanliness is an important characteristic of diesel fuel. Fuel should not contain any foreign substances, otherwise, fuel pump and injectors will have poor performance moisturizer. Moisture in the fuel can also damage or cause seizure of injector parts when corrosion occurs.
7. **Cetane Number:** The principal measure of diesel fuel quality is its cetane number. A cetane number is a measure of the delay of ignition of a diesel fuel. Higher the cetane rating, the easier the engine will start and the combustion process will be smoother within the ratings specified by the engine manufacturer. It denotes the percentage by volume of cetane (chemical name Hexadecane) in a combustible mixture containing cetane and 1-methylnapthalene. Current diesel fuels have a cetane rating between 45 and 50.

3.2 ALTERNATIVE FUELS

The sources of fossils fuels are depleting and they are not renewable. At the same time the market requirements of this fuels are increasing day by day. Hence, alternative fuels are highly essential. Alternative fuels, known as non-conventional fuel, there are many materials or substances that can be used as fuels, other than fossil fuels like petrol, diesel. Some well-known alternative fuels include biodiesel, bioalcohol (methanol, ethanol, butanol), chemically stored electricity (batteries and fuel cells), hydrogen, non-fossil methane, non-fossil natural gas, vegetable oil, propane etc.,

3.2.1 Alternative Liquid Fuels

3.2.1.1 Alcohol

In recent days, Alcohols can be considered as the best alternative fuels. Methanol and ethanol are of high interest as fuels can be produced chemically or biologically. And they have characteristics which allow them to be used in internal combustion engines. The octane ratings are higher leads to less hydrocarbon emission. Also Sulphur content is less. Figure 3.2.1.1 shown Alcohol chemical bond.

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